

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of Masters of Medical Laboratory Technology Examination April 2018

MSMLT 104 Molecular Diagnostics & Applied Genetics

Date: 18/04/2018 Day: Wednesday Time: 10.00 am To 10.30 am Maximum Marks: 20

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : MSMLT- April 2018
Semester: I
Section : I
Course Code : MSMLT_____
Course Name:_____

Important Instructions:

- There are two sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
- **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
- Marks on the right side suggest the total marks of that question.
- Maximum time allotted for **SECTION – I** is 30 minutes Question paper for **SECTION – II** will be given once the **SECTION – I** is completed within the stipulated time and returned. However, Candidates who complete **SECTION – I** earlier than 30 minutes can collect main question paper for **SECTION –II** immediately after handing over the **SECTION-I** question paper with answers.
- **SECTION – I** should be attached with the main answer book at the end of the examination and should be submitted to the supervisor.

Q - I Choose the correct answer for the following questions.

10

1. Who is responsible balance of unwind DNA after helicase works on it?

- | | |
|-------------------|-----------------|
| A. DNA polymerase | B. SSB proteins |
| C. DNA primase | D. Helicase |

2. Relaxation of DNA tension in replication is done by

- | | |
|-----------------|-------------------------|
| A. DNA Ligase | B. DNA Gyrase |
| C. DNA Helicase | D. Topoisomerase I & II |

3. Dominant Molecular Marker

- | | |
|---------|---------|
| A. RAPD | B. AFLP |
|---------|---------|

- C. RFLP

D. Both A & B
4. _____ is used for labeling in hybridization procedures.

A. Primer

B. Probe

C. Both A & B

D. None of the above
5. Termination of Transcription is done by _____ .

A. Rho protein is a transcription termination factor.

B. Rho independent transcription can also occur.

C. Rho protein is a specialized helicase

D. All of these.
6. Transcription leads to formation of _____ .

A. DNA

B. DNA & RNA

C. mRNA

D. mt DNA
7. Replication initiated by _____ .

A. DNA Polymerase

B. RNA Polymerase

C. Primase

D. Both A & B
8. Spindle fiber formation is can be stopped to study the chromosome in which phase of Mitosis

A. Prophase

B. Telophase

C. Metaphase

D. None of them.
9. Which of the following proteins is a death receptor which triggers the extrinsic pathway of apoptosis?

A. Caspases 8

B. BCL

C. FADD

D. FAS
10. In Extrinsic apoptotic pathway the Cytochrome C binds with which factor to initiate Caspase Cascade

A. APAF Protein

B. BAD

C. BCL-2

D. Cytochrome C.

Q - I (B) Match the following column “A” with Column “B”.

5

Column A	Answer	Column B
1. CFTR		A. 1 in 3000 patients
2. RAPD		B. Phylogenetic Analysis
3. Δ- Thalassemia		C. Skips Generation
4. Autosomal recessive		D. Restriction Digestion
5. RFLP		E. 1/1700

Q - I (B) Select whether following statement is true or false.

5

True/False

- 1. Accidentally Cells Die by apoptosis**
- 2. Human being follows conservative replication model**
- 3. Pedigree analysis is used to check the transmission of diseases**
- 4. Cystic Fibrosis is an autosomal recessive disorder**
- 5. Okazaki fragments are synthesized by DNA polymerase I**

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

I Semester of Masters of Medical Laboratory Technology Examination December 2017

MSMLT 104 Molecular Diagnostics & Applied Genetics

Date: 18/04/2018 Day: Wednesday Time: 10.30 am To 01.00 pm Maximum Marks: 50

Instructions:

1. There are two Sections in the paper. **SECTION –I** contains 20 Questions and **SECTION – II** contains 3 Questions.
2. **SECTION –I** should be answered in the question paper itself & **SECTION – II** should be answered in separate answer book.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **SECTION – II** is 2 Hr 30 min.
5. Draw the figure where necessary.

SECTION –II

Q - II Answer the following questions as directed. (Any Ten) 20

1. What is the application of Hardy Weinberg's Equilibrium?
2. What is Hemoglobinopathies?
3. What is Thalassemia?
4. Explain Hemolytic Disease of Newborn?
5. Explain Selection ?
6. Describe Down Syndrome?
7. What is Allelic Frequency & Genotype Frequency?
8. Explain Function of Helicases?
9. Schematically Explain Central Dogma?
10. Which class of enzyme releases the tension in replication, classify with function?
11. Differentiate between Germ Cell Division & Somatic Cell Division?

Q-III Answer the following questions as directed (Any Four) 20

1. Explain Molecular Marker & RFLP in Detail
2. Mitosis.
3. Cystic Fibrosis.
4. Explain $p^2 + 2pq + q^2 = 1$, and significance in population.
5. Explain PCR and its reactions in detail.

Q-IV Answer the following questions as directed (Any One) 10

1. Explain Intrinsic & Extrinsic Pathways in detail.
2. Explain the use of Molecular Diagnosis in Healthcare Business.

